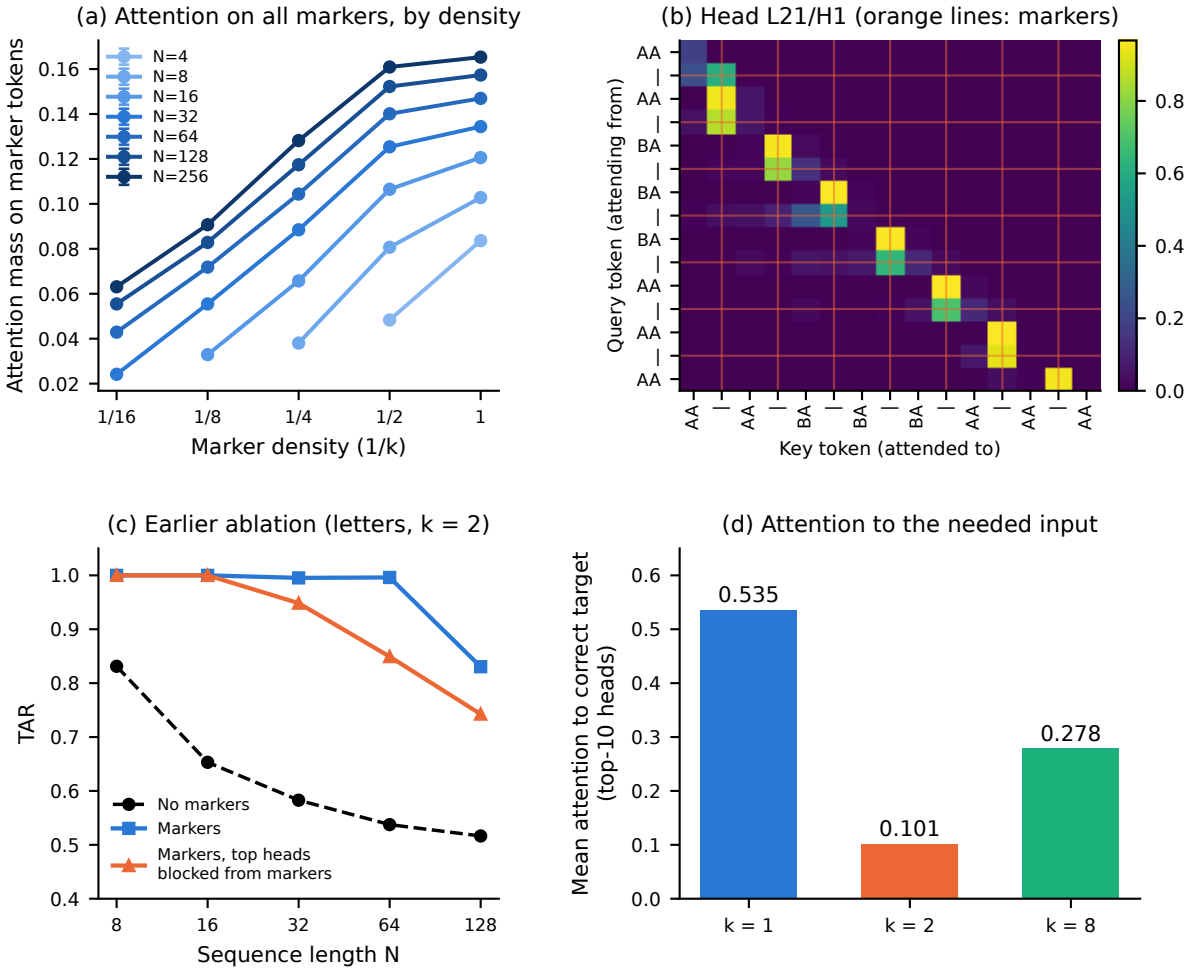
(a) Attention on all markers, by density
Attention mass on marker tokens
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
N=4
N=8
N=16
N=32
N=64
N=128
N=256
1/16
1/8
1/4
1/2
1
Marker density (1/k)
(b) Head L21/H1 (orange lines: markers)
Query token (attending from)
AA
—
AA
—
BA
—
BA
—
BA
—
AA
—
AA
—
AA
Key token (attended to)
AA
—
AA
—
BA
—
BA
—
BA
—
AA
—
AA
—
AA
0.0
0.2
0.4
0.6
0.8
(c) Earlier ablation (letters, k = 2)
TAR
0.4
0.5
0.6
0.7
0.8
0.9
1.0
No markers
Markers
Markers, top heads
blocked from markers
8
16
32
64
128
Sequence length N
(d) Attention to the needed input
Mean attention to correct target
(top-10 heads)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.535
0.101
0.278
k = 1
k = 2
k = 8